

# Introduction to Robotics, Lecture 3

Differential Kinematics

Joint and tool velocities

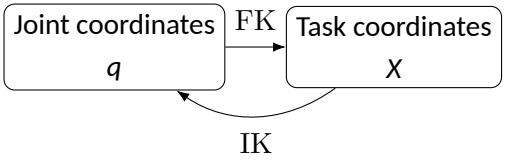
Jacobians and inverse differential maps

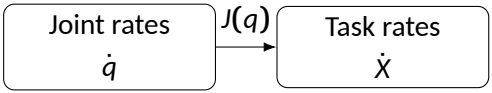
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# Where This Lecture Starts

*Pose maps are nonlinear; the velocity map is linear at a frozen configuration*




$$\dot{x} = J(q)\dot{q}$$

At a fixed $q$, the map is linear in $\dot{q}$; moving $q$ changes the map itself.

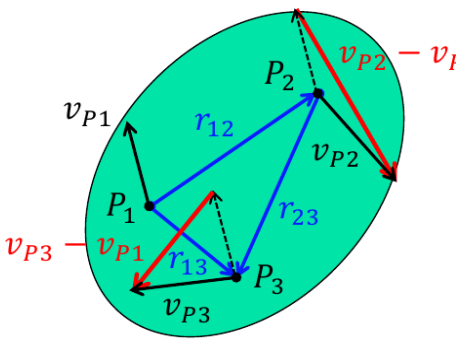
# What Differential Kinematics Is

*Instantaneous motion connects joint-space rates to linear and angular task motion*

- Differential kinematics relates joint motion to end-effector linear and angular motion.
- The mapping is instantaneous: it is evaluated at one configuration, not over an entire path.
- Differentiate direct kinematics in time, or construct the map geometrically from joint axes.
- Rotation is the subtle part: coordinate rates and angular velocity are not generally the same quantity.

# Angular Velocity of a Rigid Body

*Rigidity makes one angular velocity serve every point of the same body*




Every inter-point distance is fixed:

$$\|r_{ij}\| = \text{const} \quad \Rightarrow \quad v_{P_i} - v_{P_j} \perp r_{ij}.$$

Writing this condition for three points forces the pairwise angular rates to agree:

$$\omega_1 = \omega_2 = \omega.$$

A rigid body has one angular velocity, not one for every pair of points.

# The Fundamental Kinematic Equation

*Knowing one point velocity and the body angular velocity determines any other point*

$$v_{P_j} = v_{P_i} + \omega \times r_{ij} = v_{P_i} + S(\omega) r_{ij}, \qquad \dot{r}_{ij} = \omega \times r_{ij}.$$

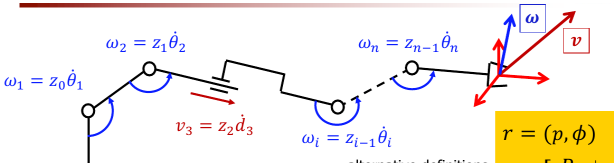
**Two limiting motions**

If $\omega = 0$, every point has the same velocity: pure translation. If two points are at rest, the body rotates about the line joining them.

The matrix $S(\omega)$ is skew-symmetric and represents cross product with $\omega$.

# One Joint, One Velocity Contribution

*Revolute and prismatic joints contribute along the axis of their preceding frame*




$$\omega_i = z_{i-1}\dot{\theta}_i \qquad v_i = z_{i-1}\dot{d}_i$$

A revolute joint contributes angular velocity; a prismatic joint contributes linear velocity. Contributions add because $v$ and $\omega$ are vectors.

# Why $\omega$ Is Not the Derivative of an Orientation

*Angular velocity is a vector, whereas a minimal orientation triple is a coordinate choice*

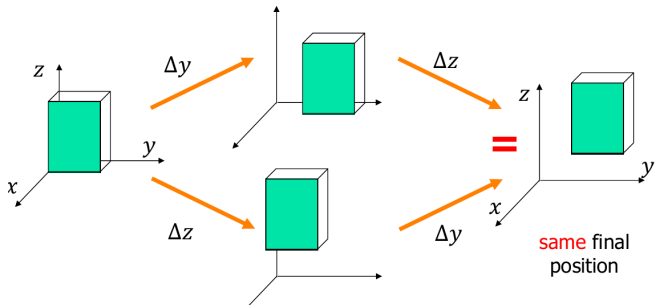
- Linear and angular velocities add in any order.
- A minimal orientation triple $\phi$ is not a vector: composing rotations is not addition of triples.
- Therefore, in general,

$$\omega \neq \dot{\phi}.$$

- Both $(p, \phi)$ and a rotation matrix $R$ are valid orientation descriptions; their rates need different treatment.

# Translations Always Commute

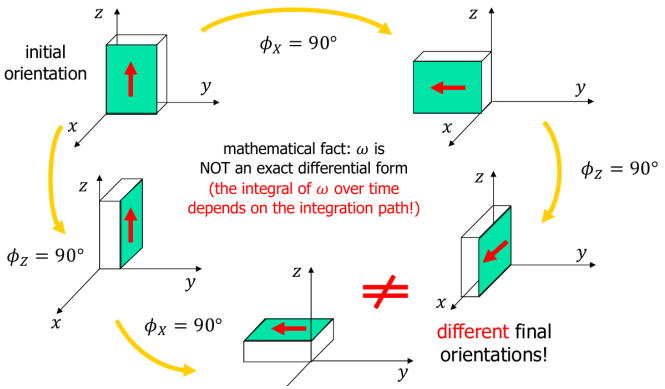
*The order of two displacements does not affect the final position*




$\Delta y$ then $\Delta z = \Delta z$ then $\Delta y$. Finite and infinitesimal displacements both add as vectors; no ordering bookkeeping is needed.

# Finite Rotations Do Not Commute

*Changing the order of rotations about different axes changes the final orientation*




$$R_X(\pi/2)R_Z(\pi/2) \neq R_Z(\pi/2)R_X(\pi/2).$$

Rotations about one fixed axis are the special commuting case.

# $\omega$ Is Not an Exact Differential

*Equal axis-wise areas of angular-rate profiles can lead to different orientations*

Two rate profiles can satisfy

$$\int_0^T \omega \, dt$$

with the same vector result while producing different final rotation matrices $R_f$. The integral depends on the path through orientation space.

Coordinate rates are different:

$$\int_0^T \dot{\phi} \, dt = \phi_f - \phi_i.$$

This telescopes because $\dot{\phi}$ is an exact differential of its chosen coordinates.

# Infinitesimal Rotations Do Commute

*Keeping only first-order terms recovers an additive differential description*

$$R(d\phi) = I + S(d\phi), \qquad S(d\phi) = \begin{bmatrix} 0 & -d\phi_Z & d\phi_Y \\ d\phi_Z & 0 & -d\phi_X \\ -d\phi_Y & d\phi_X & 0 \end{bmatrix}.$$

Multiplying elementary infinitesimal rotations in any order gives the same first-order matrix. Products of two small increments are discarded, not assumed to be zero.

# Time Derivative of a Rotation Matrix

*A matrix derivative reveals angular velocity without choosing orientation angles*

Orthogonality gives a skew-symmetric product:

$$\dot{R}R^T = S, \qquad S = S(\omega).$$

Hence

$$\dot{R} = S(\omega)R \qquad \Longleftrightarrow \qquad S(\omega) = \dot{R}R^T.$$

This is the matrix form of angular velocity and will later extract the angular part of a Jacobian column.

# Worked Example: Differentiating an Elementary Rotation

*For a rotation about* **x**, *the matrix identity returns the expected axis velocity*

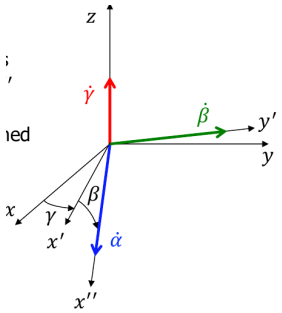
$$R_X(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \quad \Rightarrow \quad \dot{R}_X R_X^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\phi} \\ 0 & \dot{\phi} & 0 \end{bmatrix}.$$

$$\omega = \dot{\phi} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

More generally, an axis–angle rotation $R(r, \theta)$ has $\omega = \dot{\theta} r$ when its axis is fixed.

# RPY Rates Against Angular Velocity

*A coordinate singularity, not a robot singularity*




The Z, new-Y, and newer-X rotation axes form the columns of $T_{\mathrm{RPY}}$:

$$\omega = T_{\mathrm{RPY}}(\beta, \gamma) \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix}.$$

Its determinant is $\det T_{\mathrm{RPY}} = \cos\beta$. At $\beta = \pm\pi/2$, this representation cannot express every angular velocity by angle rates.

# Two Jacobians, Not One

*Analytic and geometric Jacobians answer different instantaneous questions*

## Analytic Jacobian

For task coordinates $r = f_r(q)$,

$$\dot{r} = \frac{\partial f_r}{\partial q}\dot{q} = J_r(q)\dot{q}.$$

It is obtained by differentiating the task map.

## Geometric Jacobian

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} J_L(q) \\ J_A(q) \end{bmatrix} \dot{q} = J(q)\dot{q}.$$

It maps to the physical end-effector twist.

Their lower halves differ because $\omega \neq \dot{\phi}$ in general.

# The Jacobian Equation

*Each entry is a local sensitivity, and the matrix changes with configuration*

$$\dot{X}_{6\times 1} = J_{6\times n}(q)\dot{q}_{n\times 1}, \qquad J = \left[\frac{\partial f_i}{\partial q_j}\right]_{\substack{i=1,\ldots,6\\ j=1,\ldots,n}}.$$

- A six-quantity task and a seven-joint robot give a $6 \times 7$ Jacobian.
- $J(q)$ is recomputed at every configuration; it is not a constant robot property.
- Column $j$ records the effect of moving only joint $j$ at that configuration.

# Which Vector Is on the Left?

*Coordinate rates use the analytic Jacobian; physical twist uses the geometric Jacobian*

**Coordinate-rate output**

$$\begin{bmatrix} \dot{p} \\ \dot{\phi} \end{bmatrix} = J_r(q)\dot{q}.$$

The partial-derivative array is correct for the stack $(\dot{p}, \dot{\phi})$.

**Physical-twist output**

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q}.$$

Here $\omega$ is not the derivative of three orientation coordinates.

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \mathrm{blkdiag}(I, T(\phi)) \begin{bmatrix} \dot{p} \\ \dot{\phi} \end{bmatrix}.$$

# A Jacobian Is a Linear Map Between Two Spaces

*The same configuration-dependent map appears in velocity, statics, and stiffness models*

**Motion**

Small joint displacements map to small Cartesian displacements.

**Statics**

$$\tau = J^T(q)F.$$

Forces and moments map back to joint torques.

**Stiffness**

Joint and Cartesian stiffness are related locally through the same geometry.

The underlying kinematics is nonlinear, but its local linear map is enough for small changes at one $q$.

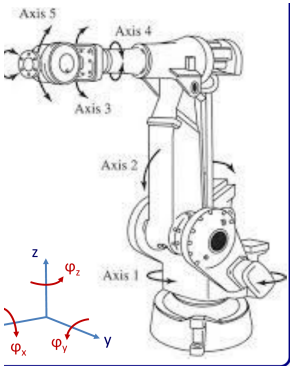
# Physical Meaning of One Column

*A column asks how the pose changes when only one joint moves*

For the vertical base axis in the illustrated arm,

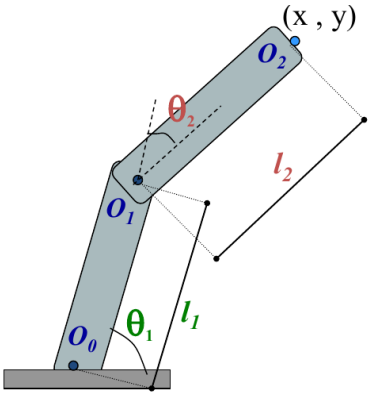
$$J_1 = \begin{bmatrix} -(L_3 + L_6)\sin q_1 \\ (L_3 + L_6)\cos q_1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

It sweeps the end effector in $x$ and $y$, gives no vertical motion or tool tilt, and turns the tool about $z$.



# Classical Approach: Differentiate the Direct Kinematics

*For the planar 2R arm, the task coordinates are explicit functions of two angles*




$$x = l_1 c_1 + l_2 c_{12}, \qquad y = l_1 s_1 + l_2 s_{12},$$

$$\varphi = \theta_1 + \theta_2, \qquad J = \frac{\partial(x, y, \varphi)}{\partial(\theta_1, \theta_2)}.$$

Differentiating column one means following both links; differentiating column two follows the second link only.

# The 2R Analytic Jacobian, Evaluated

*The chain rule makes the first column contain the contribution of both links*

$$J = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \\ 1 & 1 \end{bmatrix}.$$

**Read the columns physically**

Changing $\theta_1$ moves both links, whereas changing $\theta_2$ moves only link 2. The last row is all ones because the tool angle is the sum of the two joint angles.

# Geometric Rule: Columns from Joint Axes

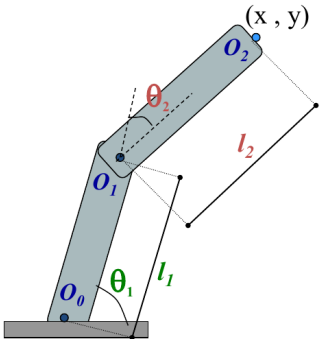
*A column can be built directly from the relevant axis and the tool moment arm*

$$J_i = \begin{cases} \begin{bmatrix} z_{i-1} \times (o_n - o_{i-1}) \\ z_{i-1} \end{bmatrix}, & \text{revolute joint,} \\ \begin{bmatrix} z_{i-1} \\ 0 \end{bmatrix}, & \text{prismatic joint.} \end{cases}$$

Here $z_{i-1}$ is the third column and $o_{i-1}$ the translation column of ${}^{0}T_{i-1}$. A prismatic joint has no angular contribution; a revolute joint has both a moment-arm velocity and an angular-axis velocity.

# Setting Up the 2R Arm for the Geometric Rule

*Planar joint axes and origins supply the moment arms*




$$o_0 = 0, \quad o_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix},$$

$$o_2 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ 0 \end{bmatrix}.$$

$$z_0 = z_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

# Column One by Cross Product

*Crossing the vertical base axis with the tool position rotates the planar moment arm*

$$z_0 \times (o_2 - o_0) = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} \\ 0 \end{bmatrix}.$$

$$J_1 = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

The lower half is the joint axis itself, so this planar joint contributes only $\omega_z$.

# Column Two, and the Two Routes Agree

*The elbow moment arm contains only link 2, giving the second column directly*

$$J_2 = \begin{bmatrix} -a_2 s_{12} \\ a_2 c_{12} \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Its moment arm begins at the elbow origin, so link 1 does not appear.

Deleting the spatial rows this arm cannot move leaves

$$\begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} \\ 1 & 1 \end{bmatrix}.$$

It agrees entry for entry with the differentiated result.

# Differentiating the Chain Instead of the Equations

*Replace only the elementary factor that contains the selected joint variable*

$$T = T_{\text{base}}\,[\,T_1(q)\,H(q_i)\,T_2(q)\,]\,T_{\text{tool}}.$$
$$T'_{q_i} = T_{\text{base}}\,[\,T_1(q)\,H'(q_i)\,T_2(q)\,]\,T_{\text{tool}}.$$

Every other factor is untouched because it does not depend on $q_i$. One such product per joint gives all derivative matrices; this is the chain rule arranged for an implementation loop.

# Reading a Jacobian Column out of the Derivative Matrix

*Translation comes from the last column; angular velocity comes from a skew matrix*

$$[J]_{1..3,i} = [T'_{q_i}]_{1..3,4}.$$

To get the angular half, remove the current orientation:

$$S(\omega) = T'_{q_i} T^{-1}$$

in the upper-left $3 \times 3$ block. Read the vector as

$$\omega_x = S_{32}, \qquad \omega_y = S_{13}, \qquad \omega_z = S_{21}.$$

# Elementary Translations and Their Derivatives

*A prismatic elementary transform differentiates to one fixed matrix entry*

$$\frac{\partial T_x}{\partial x} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \frac{\partial T_y}{\partial y} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \frac{\partial T_z}{\partial z} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The derivative is independent of the translation value: it simply selects the corresponding coordinate in the last column.

# Elementary Rotations and Their Derivatives

*At zero angle, rotation derivatives are the embedded skew generators*

$$R_x'(0) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad R_y'(0) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad R_z'(0) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

In homogeneous-transform form, add a zero fourth row and column. These are exactly the skew generators from the infinitesimal-rotation construction; zero angle is a special case, not the general derivative.

# Three Routes, One Jacobian

*Choose the construction that fits the available model and the task at hand*

| Task equations | Joint axes | Transform chain |
| --- | --- | --- |
| Differentiate closed-form kinematics. Shortest on paper, but requires explicit task equations. | Build columns from axes and moment arms. No symbolic differentiation is needed. | Replace an elementary factor by its derivative. Mechanical and suitable for code. |

The first route gives an analytic Jacobian; the geometric and transform-chain routes give the geometric Jacobian. At the same configuration they are related by the representation-rate map.

# The Inverse Differential Problem

*Given an end-effector velocity, find joint rates that realize it*

$$v = J(q)\dot{q} \qquad \Rightarrow \qquad \dot{q} = J^{-1}(q)v$$

only when $J$ is square and nonsingular. Two failures are fundamental:

- near a singularity, an inverse may yield physically excessive joint rates;
- for a redundant robot, $J$ is rectangular and has no ordinary inverse.

# Inversion as a Minimum-Norm Problem

*The pseudoinverse chooses the smallest joint-rate vector among the available solutions*

$$\min \frac{1}{2}\|\dot{q}\|^2 \qquad \text{subject to} \qquad J\dot{q} = v.$$

Its solution is

$$\dot{q} = J^{\#} v.$$

If the demand is infeasible, the same construction first minimizes the task-space error and then chooses the smallest-norm joint vector among those minimizers.

# The Pseudoinverse

*One unique matrix always exists; convenient product formulas require full rank*

$$JJ^{\#}J = J, \quad J^{\#}JJ^{\#} = J^{\#}, \quad (JJ^{\#})^T = JJ^{\#}, \quad (J^{\#}J)^T = J^{\#}J.$$

| Square, full rank | Full row rank | Full column rank |
|---|---|---|
| $J^{\#} = J^{-1}$ | $J^{\#} = J^T(JJ^T)^{-1}$ | $J^{\#} = (J^TJ)^{-1}J^T$ |

At rank deficiency, compute it from a singular-value decomposition rather than these products.

# A Singular 2R Arm, Worked

*At full extension the two Jacobian columns are parallel and the rank is one*

$$J = \begin{bmatrix} -2s_1 & -s_1 \\ 2c_1 & c_1 \end{bmatrix}, \qquad J^{\#} = \frac{1}{5}\begin{bmatrix} -2s_1 & 2c_1 \\ -s_1 & c_1 \end{bmatrix}.$$

Neither full-rank product formula applies. Instead, $JJ^{\#}$ and $J^{\#}J$ are symmetric and the four Penrose relations hold.

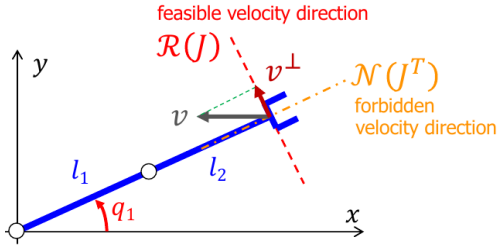
# Feasible and Forbidden Directions

*At a singular configuration, the pseudoinverse projects a requested velocity onto the range of J*

$$J\dot{q} = JJ^{\#}v = v^{\perp}.$$

The range of $J$ contains feasible task velocities; its orthogonal complement contains directions that no joint rate can produce.

$$q_1 = \pi/6: \quad \dot{q} = \begin{bmatrix} 0.1 \\ 0.05 \end{bmatrix}, \quad v^{\perp} = \begin{bmatrix} -1/8 \\ \sqrt{3}/8 \end{bmatrix}.$$



# Inverting at Higher Differential Order

*Differentiating the velocity relation brings in the Jacobian rate and acceleration*

$$\ddot{q} = J_r^{-1}(q)\left(\ddot{r} - \dot{J}_r(q)\dot{q}\right).$$

One more product rule gives

$$\dddot{q} = J_r^{-1}(q)\left(\dddot{r} - 2\dot{J}_r(q)\ddot{q} - \ddot{J}_r(q)\dot{q}\right).$$

The inverse, or a pseudoinverse when necessary, remains the leading map at each differential order. Integrating a higher-order command down produces smoother joint motion.

# What This Leaves Open

*The Jacobian exposes the next control and modelling questions*

- How can singular configurations be found and avoided?
- What should a redundant arm do with its unused joint-rate freedom?
- How does the statics duality map forces to joint torques?
- How should numerical conditioning near singularities be handled, including damped alternatives?

# Reference Materials

- A. Klimchik — L3 Differential kinematics v01
- A. De Luca — 11 DifferentialKinematics
- A. De Luca — 12a InverseDifferentialKinematics

# Deserve “A” grade!

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